

DATA POISONING TO FAKE A NASH EQUILIBRIUM IN MARKOV GAMES

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Markov Game

- A finite-horizon n -player Markov game $\mathcal{G} = (\mathcal{S}, \mathcal{A}, P, R, T)$ is:
 - $\mathcal{S}$ is the finite state space,
 - $\mathcal{A} = \mathcal{A}_1 \times \mathcal{A}_2$ is the finite joint action space,
 - $P_t : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$ for $t \in [T]$ is the transition kernel with initial state distribution,
 - $R_t : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^n$ for $t \in [T]$ is the mean reward function,
 - T is the finite horizon,

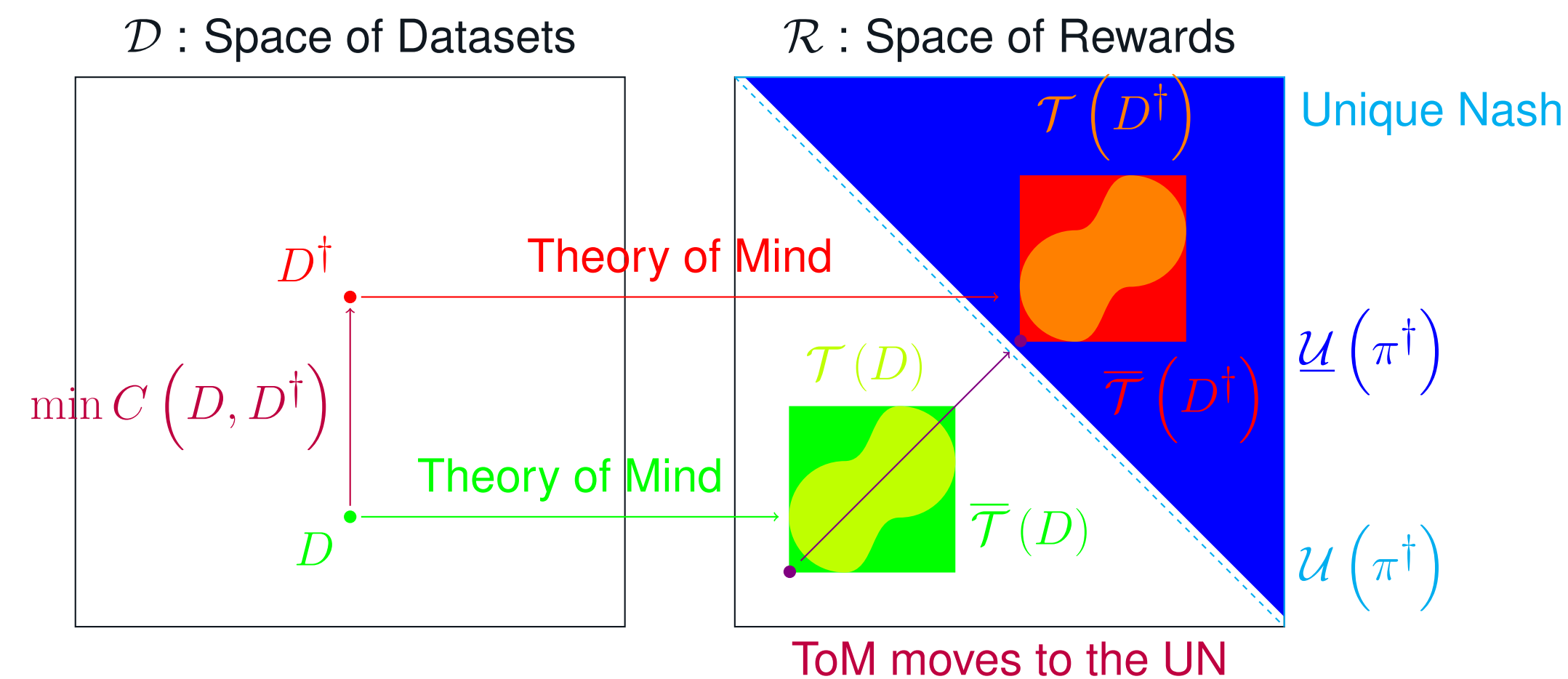
the Victims

- The victims are uncertainty-aware offline learners that use additive bonus terms β when estimating the Q function, for example, $Q = \hat{R} - \beta + \mathbb{E}_{\hat{P}}[V']$.
- The victims learn a two-player zero-sum Markov game from a training set $\left\{ \left(\left(s_t^{(k)}, a_t^{(k)}, r_t^{(k)} \right)_{t=1}^T \right) \right\}_{k=1}^K$, with $r_t^{(k)} \in [-b, b]$ with K episodes.

the Attacker

- The attacker wants the victims to learn a target (deterministic) policy $\pi^\dagger$ as the unique Markov perfect (Nash) equilibrium.
- The attacker can modify the rewards in the training set from r^o to $r^\dagger$.
- The attacker minimizes the reward modification cost $\left\| r^\dagger - r^o \right\|_1$, for example, $\sum_{k=1}^K \sum_{t=1}^T \left\| r_t^{\dagger, (k)} - r_t^{o, (k)} \right\|_1$.
- The attacker does not know $\hat{R}$ and $\hat{P}$, but has theory-of-mind about the victims satisfying $\left| \hat{R} - R^{(\text{MLE})} \right| < \rho^{(R)}$ and $\left\| \hat{P} - P^{(\text{MLE})} \right\|_1 < \rho^{(P)}$.

the Attack Problem



the Attack Formulation

- The attack can be formulated as

$$\min_{r^\dagger} \left\| r^\dagger - r^o \right\|_1$$

$$\text{subject to } \mathcal{T} \left(r^\dagger; \rho^{(R)}, \rho^{(P)} \right) \subseteq \mathcal{N}^{-1} \left(\{ \pi^\dagger \} \right),$$

where,

- $\mathcal{T}(r)$ is the theory-of-mind (TOM) set of the attacker containing the set of plausible Q functions computed based on reward data r evaluated on $\pi^\dagger$,
- $\mathcal{N}^{-1}(\{\pi\})$ is the (inverse) unique Nash (UN) polytope of Q functions such that π is the (unique) strict Markov perfect (Nash) equilibrium.



Feasibility

- The attack is feasible if $\rho_t^{(R)}(s, a) + |\beta_t(s, a)| < \frac{b - \iota}{4T}, \forall t, s$, and actions a such that $a_1 = \pi_{1,t}^\dagger(s)$ or $a_2 = \pi_{2,t}^\dagger(s)$.
- For example, if $\rho^{(R)} = 0$ and $\beta = \frac{c}{\sqrt{N_t(s, a)}}$, then the condition is a data coverage condition, $N_t(s, a) > 16cT^2$ for actions profiles in the same row or column as $\pi^\dagger$ in the stage game matrices.
- Suppose $\pi^\dagger = (1, 1)$ in a stage game, then the following attack is feasible under the previous feasibility condition,

$\mathcal{A}_1 \setminus \mathcal{A}_2$	1	2	3	4
1	0.5	1	1	1
2	0	-	-	-
3	0	-	-	-
4	0	-	-	-

- Unspecified cells' corresponding rewards do not need to be poisoned.

Linear Program

- The following attacker's problem can be converted into a linear program.

$$\min_{r^\dagger} \sum_{k=1}^K \sum_{t=1}^T \left\| r_t^{\dagger, (k)} - r_t^{o, (k)} \right\|_1$$

$$\text{subject to } \overline{\mathcal{T}} \left(r^\dagger; \rho^{(R)}, \rho^{(P)} \right) \subseteq \mathcal{N}^{-1} \left(\{ \pi^\dagger \}; \iota \right),$$

where,

- $\overline{\mathcal{T}}(r)$ is an outer approximation of the theory-of-mind (TOM) set that is a polytope and can be characterized by linear constraints,
- $\mathcal{N}^{-1}(\{\pi\}; \iota)$ is a closed subset of the (inverse) unique Nash (UN) polytope with ι gaps to avoid strict inequalities in the constraints.

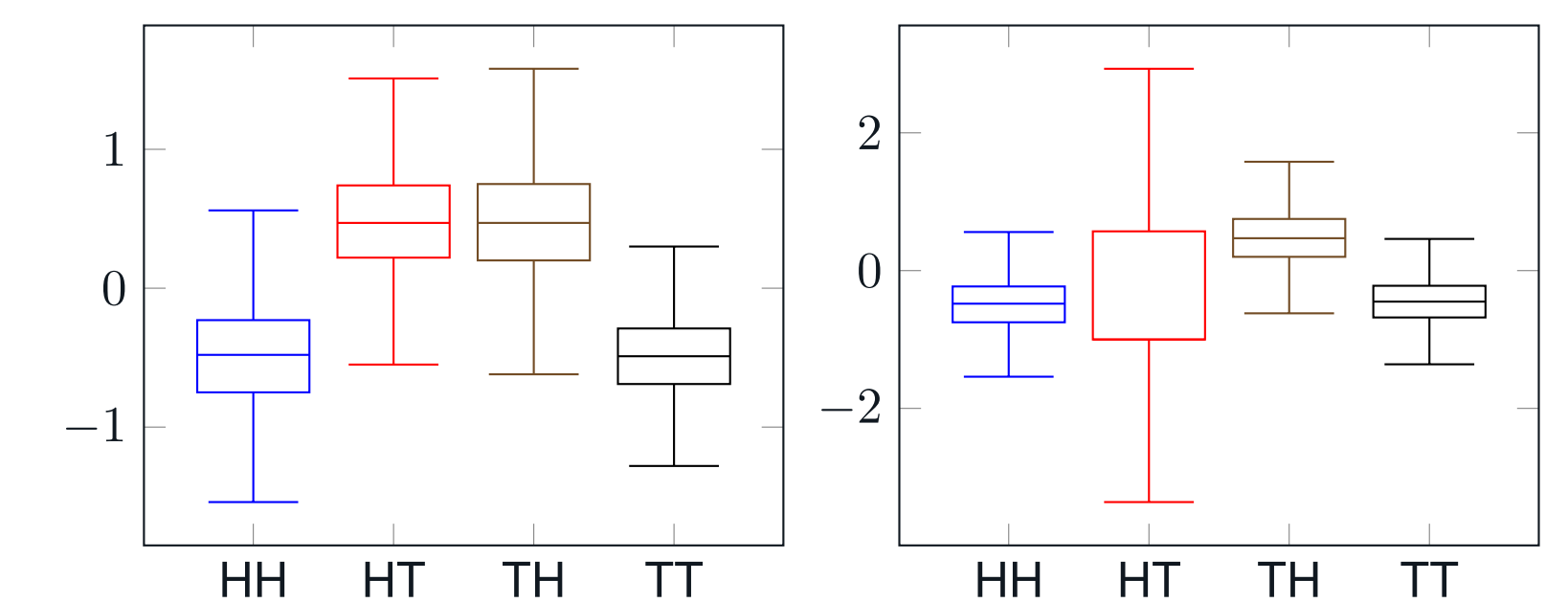
Experiments

- Rock Paper Scissors game: original (left) and poisoned (right) with $\pi^\dagger = (R, R)$, ($\iota = 0.01$ and "-" means unchanged).

	R	P	S
R	0	-1	1
P	1	0	-1
S	-1	1	0

	R	P	S
R	0	0.01	1
P	-0.01	-	-
S	-1	-	-

- Stochastic Matching Penny: original (left) and poisoned (right) with $\pi^\dagger = (H, H)$, and the plots are showing distributions of rewards.



Average costs	$n = 1$	$n = 10$	$n = 100$
Our attack	1.06	9.09	99.47
Feasible attack	2.12	16.08	250.46
DSE attack	2.06	18.31	198.38

Summary

- We characterize offline data poisoning attacks on Multi-Agent Reinforcement Learning (MARL), where an attacker may change a data set in an attempt to install a (potentially fictitious) unique Markov-perfect Nash equilibrium for a two-player zero-sum Markov game.
- Our work sheds light on the structure of data poisoning attacks on offline MARL, a necessary step before one can design more robust MARL algorithms.